

3.2 解:

自感式传感器电感值表达式: $L_0 = \frac{W^2 \mu_0 S_0}{2l_0}$

$$\text{对单极式, 有 } \Delta L = L_1 - L_0 = \frac{W^2 \mu_0 S_0}{2(l_0 + \Delta l)} - \frac{W^2 \mu_0 S_0}{2l_0}$$

$$= \frac{W^2 \mu_0 S_0}{2l_0} \left(\frac{l_0}{l_0 + \Delta l} - 1 \right)$$

$$S_1 = \frac{\Delta L}{\Delta l} = -\frac{l_0}{l_0} \left[1 - \frac{\Delta l}{l_0} + \left(\frac{\Delta l}{l_0} \right)^2 + \dots \right]$$

$$\text{对差分式, 有 } \Delta L_2 = L_2 - 2L_0 = 2(L_1 - L_0) = \frac{W^2 \mu_0 S_0}{l_0} \left(\frac{l_0}{l_0 + \Delta l} - 1 \right)$$

$$S_2 = \frac{\Delta L_2}{\Delta l} = -\frac{2l_0}{l_0} \left[1 - \frac{\Delta l}{l_0} + \left(\frac{\Delta l}{l_0} \right)^2 + \dots \right]$$

$S_2 = 2S_1$, 差分式与单极式相比其灵敏度提高了一倍, 非线性得到减小

$$3.5 \text{ 解: } \text{记 } l_1 = 60 \text{ mm} \quad S_1 = 30 \times 10^{-6} \text{ m}^2 = 300 \text{ mm}^2 \quad l_2 = 10 \text{ mm} \quad S_2 = 15 \times 30 \text{ mm}^2 = 450 \text{ mm}^2$$

$$R_1 = \frac{l_1}{\mu_1 S_1} + \frac{l_2}{\mu_1 S_2} + \frac{l_3}{\mu_1 S_3}$$

$$l_3 = 60 \text{ mm} \quad S_3 = 2 \times 30 \text{ mm}^2 = 60 \text{ mm}^2$$

$$\mu_1 = 4\pi \times 10^{-3} \text{ H} \cdot \text{m}^{-1}$$

$$\text{记 } l_1 = 60 - 15 \text{ mm} = 0.045 \text{ m}$$

$$S_1 = 0.03 \times 0.01 \text{ m}^2$$

$$l_2 = 2 \left(20 - \frac{10}{2} \right) \text{ mm} = 0.03$$

$$S_2 = 0.015 \times 0.03 \text{ m}^2$$

$$2 \left(20 - \frac{10}{2} + \frac{2}{2} \right) \text{ mm} = 0.032$$

$$S_3 = 0.03 \times 0.002 \text{ m}^2$$

$$\mu_1 = 4\pi \times 10^{-3} \text{ H} \cdot \text{m}^{-1}$$

$$\text{铁芯磁阻 } R_1 = \frac{l_1}{\mu_1 S_1} + \frac{l_2}{\mu_1 S_2} + \frac{l_3}{\mu_1 S_3}$$

$$= \left(\frac{0.045}{4\pi \times 10^{-3} \times 3 \times 10^{-4}} + \frac{0.032}{4\pi \times 10^{-3} \times 4.5 \times 10^{-4}} + \frac{0.045}{4\pi \times 10^{-3} \times 6 \times 10^{-5}} \right) \text{ H}^{-1}$$

$$= 7.73 \times 10^4 \text{ H}^{-1}$$

空气磁阻: 当气隙为 0 时 $R_{21} = 0$

$$\text{当气隙为 } 2 \text{ mm 时 } R_{22} = 2 \times \frac{0.002}{4\pi \times 10^{-7} \times 0.03 \times 0.015} = 7.07 \times 10^3 \text{ H}^{-1}$$

$$\text{空气气隙为 0 时 } L = \frac{W^2}{R_1} = 0.517 \text{ H}$$

$$\text{空气气隙为 } 2 \text{ mm 时 } L_2 = \frac{W^2}{R_1 + R_{22}} = 0.474 \text{ H}$$

3.6 解:

设 $y = bx + a$

X	0.3	0.5	1.0	1.5	2.0	3.0	4.0	5.0	6.0
f	2.523	2.502	2.461	2.432	2.410	2.380	2.362	2.351	2.343
y	-1.66	-1.78	-2.06	-2.31	-2.56	-3.06	-3.54	-4.02	-4.61

最小二乘法拟合直线:

$$b = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$$a = \frac{\sum x_i^2 \sum y_i - \sum x_i \cdot \sum x_i y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$$\sum x_i y_i = -83.133 \quad \sum x_i = 23.3 \quad \sum y_i = -25.6$$

$$\sum x_i^2 = 93.59$$

$$n = 9$$

$$\text{代入得} \begin{cases} b = -0.51 \\ a = -1.53 \end{cases}$$

$$\text{则 } y = -0.51x - 1.53$$

$$\Delta_1 = 0.023 \quad \Delta_2 = 0.005 \quad \Delta_3 = 0.02 \quad \Delta_4 = 0.015$$

$$\Delta_5 = 0.01 \quad \Delta_6 = 0 \quad \Delta_7 = 0.03 \quad \Delta_8 = 0.06 \quad \Delta_9 = 0.02$$

$$\Delta L_{\max} = \Delta_8 = 0.06 \quad y_{FS} = -4.61$$

$$Y_2 = 1 \frac{\Delta L_{\max}}{y_{FS}} \times 100\% = \pm 1.30\%$$

工作特性方程: $f = e^{(-0.51x - 1.53)} + 2.333$